MAHESH REDDY

Andhra Pradesh, India | +91 91773 19640 | maheshreddygit@gmail.com
LinkedIn | GitHub | Portfolio | Blog

PROFESSIONAL SUMMARY

B.Tech Computer Science (AI & Data Science) undergraduate (Class of 2027) with hands-on experience building machine learning systems across NLP, computer vision, and classification, achieving 92–99% model accuracy across projects. GSSoC'26 open-source contributor with merged pull requests to production ML repositories. First author of a research project on Emotionally Aware AI at Parul University, and creator of myMLlib, an open-source ML library. Proficient in Python, PyTorch, TensorFlow, and scikit-learn, seeking AI/ML research internship opportunities.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science (Artificial Intelligence & Data Science) **Expected May 2027**
Parul University, Gujarat, India | Dept. of AI and Data Science
Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, Natural Language Processing (NLP), Computer Vision, Data Structures and Algorithms, OOP, DBMS

EXPERIENCE

GSSoC'26 Open Source Contributor **May 2026 – Present**
GirlScript Foundation | Open Source Program

- Contributed multiple merged pull requests across 3+ production ML repositories, improving preprocessing modules, model training pipelines, and classification systems through code review and issue-tracking workflows; recognized with the GitHub Pull Shark achievement for sustained open-source contributions.

ACHIEVEMENTS

- Nothing Playground Hackathon (Nothing) – Selected to the Top 50 nationally for an original creation submitted to Nothing's Playground Hackathon.
- Vadodara Hackathon 6.0 – Placed in the Top 100 within Parul University's internal round.

PROJECTS

MiniGPT-Emotional-Support V2 **2026**
Python, PyTorch, Transformer Architecture, Byte Pair Encoding (BPE) Tokenizer, Natural Language Processing (NLP) | GitHub Repo

- Built a from-scratch GPT-style transformer (RMSNorm, SwiGLU, weight tying) to test whether a small, locally trainable model could learn genuine empathetic response patterns; trained on 52,137 pairs across three emotional-support corpora with a custom 16k BPE tokenizer, improving internal evaluation from 4/10 (V1) to 7.5/10 (V2).

Student Performance Risk Portal **2026**
Python, scikit-learn, Pandas, REST API | GitHub Repo | Live Demo

- Built an early-warning tool so a student or advisor can spot academic risk building up before grades slip; deployed a scikit-learn classification pipeline over habit and lifestyle data behind a REST API returning risk level, confidence scores, and actionable recommendations, live on Render.

Sign Language Detection System **2026**
Python, TensorFlow, Keras, Convolutional Neural Network (CNN), Computer Vision | GitHub Repo

- Built to lower a communication barrier for hearing-impaired individuals by translating hand gestures into letters in real time; engineered a baseline CNN plus a transformer-based SignFormer model, paired with a MediaPipe hand-landmark pipeline for live webcam inference.

Customer Intelligence System **2026**
Python, TensorFlow, Keras, scikit-learn, Pandas, FastAPI | GitHub Repo

- Built to test whether combining classical clustering with a learned embedding space gives more usable customer segmentation than either alone; engineered RFM behavioral features on UCI Online Retail data, evaluated K-Means clusters via silhouette scores, and added a neural embedding layer exposed through a FastAPI backend.

LSTM Chatbot with Word2Vec Embeddings **2026**

Python, TensorFlow, Keras, Long Short-Term Memory (LSTM), Word2Vec, Natural Language Processing (NLP) | GitHub Repo

- Built as a foundational study of sequence modeling ahead of the transformer-based MiniGPT project; implemented the full pipeline from tokenization through Word2Vec embedding training to LSTM-based sequence modeling and response generation.

RESEARCH AND PUBLICATIONS

Emotionally Aware AI Companions: A Human-Centric Methodological Approach **2026**

First Author | Co-authors: Sanjay, Harish, Sindhu, Rahul Kumar Moud | Dept. of AI and Data Science, Parul University

- Proposed a modular architecture combining multimodal emotion perception, contextual memory management, and stable character modeling to address the LLM "goldfish memory" problem, formalizing a mathematical stability constraint and curriculum-learning strategy grounded in ethical AI safety priorities.

Feature Scaling and Gradient Descent in Logistic Regression: A Breast Cancer Detection Case Study **2025**

Independent Researcher, Self-Published

- Compared min-max normalization and standardization on gradient descent convergence using the Breast Cancer Wisconsin dataset (Kaggle), implementing the full training pipeline from scratch in Python/NumPy and achieving 92%+ accuracy.

TECHNICAL SKILLS

Programming Languages: Python, C, SQL

ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Transformer Architecture, CNN, RNN, LSTM, Neural Networks, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Natural Language Processing: Word2Vec, TF-IDF, BPE Tokenization, Text Classification, Sequence Modeling, Language Modeling, Generative Text Models

Computer Vision: Image Classification, Real-Time Object Detection, Image Preprocessing, Feature Extraction

AI Research: Multimodal AI, Emotion Recognition, Human-AI Interaction, Ethical AI Design, Curriculum Learning, Reinforcement Learning Concepts

Software Engineering: REST API Development (FastAPI), OOP, Data Structures and Algorithms, DBMS, Git, GitHub, Open-Source Development

Tools and Platforms: Jupyter Notebook, VS Code, Google Colab, Kaggle, Render

CERTIFICATIONS AND TECHNICAL WRITING

CS50: Introduction to Programming with Python – Harvard University (edX) | 2025

Portfolio: maheshreddyml.netlify.app

AI/ML Technical Blog: throughmyeyesaiml.blogspot.com – in-depth articles on gradient descent, from-scratch ML, NLP architecture, and ethical AI design for software developer audiences.